Personal Expense Pattern Analysis and Transaction Classification Using Machine Learnin

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Abstract

Managing personal finances requires understanding spending habits and identifying the difference between income and expenses. Machine learning can help analyze transaction records and automatically classify financial transactions. This study presents an exploratory approach to personal expense pattern analysis and transaction classification using Python and machine learning techniques. The dataset contains 277 financial transactions recorded between November 2021 and March 2022. Data preprocessing, exploratory data analysis, and visualization were performed to understand transaction patterns. A Logistic Regression classifier was developed to classify transactions into Income and Expense categories. Its performance was compared with a majority-class baseline using accuracy, precision, recall, and F1-score. The Logistic Regression model achieved an accuracy of 94.3% and a weighted F1-score of 93.8%, compared with 82.9% accuracy for the baseline. However, the dataset is small and imbalanced, with significantly more expense records than income records. Therefore, the results should be interpreted as an exploratory experiment rather than evidence of reliable performance on unseen users or future transactions.

INTRODUCTION

Personal financial management is important for understanding spending habits, controlling unnecessary expenses, and making better financial decisions. Individuals often record financial transactions containing information such as transaction dates, amounts, descriptions, and categories. Analyzing these records can help identify spending patterns and understand how money is used.

Traditional expense tracking generally requires users to examine and categorize transactions manually. This process can become time-consuming when the number of transactions increases. Machine learning offers methods for learning patterns from existing data and classifying new observations.

This project explores the application of machine learning to personal financial transaction records. It combines data cleaning, exploratory data analysis, visualization, and supervised classification.

The primary objectives are:

1. To examine personal financial transaction data.
2. To identify patterns in transaction records using tables and graphs.
3. To develop a machine learning model for classifying transactions as Income or Expense.
4. To compare the model with a simple majority-class baseline.
5. To evaluate the model using standard classification metrics.

The current implementation focuses on transaction classification and exploratory analysis. It does not claim to provide a validated forecast of future monthly expenditure.

RELATED WORK

Machine learning has been widely used for classification and predictive analysis across different application areas. Logistic Regression is a commonly used supervised learning algorithm for classification problems. It estimates class probabilities using a mathematical relationship between input features and the target class.

Text-processing techniques can also convert transaction descriptions into numerical features. Term Frequency–Inverse Document Frequency (TF-IDF) is a standard technique for representing text according to the importance of its terms within a collection of documents.

The scikit-learn library provides implementations of machine learning algorithms, data preprocessing pipelines, and evaluation metrics. These facilities support reproducible experiments and comparisons between classification models.

This project applies these established techniques to a small financial transaction dataset. Its contribution is an implementation and exploratory evaluation of a transaction-classification workflow, rather than a claim of developing a new machine learning algorithm.

DATASET DESCRIPTION

A. Dataset Overview

The experiment uses a CSV file containing personal financial transaction records. The notebook identifies 277 transactions covering the period from November 21, 2021, to March 2, 2022.

The dataset includes fields related to transaction dates, accounts, categories, transaction descriptions, amounts, and transaction types.

The target variable is **Transaction_Type**, which contains two classes:

- Expense: 231 records
- Income: 46 records

Thus, approximately 83.4% of the records belong to the Expense class, while 16.6% belong to the Income class.

The dataset is imbalanced because expense transactions substantially outnumber income transactions.

Dataset provenance: The experiment uses the CSV supplied for this assignment. The original dataset's public source and collection methodology have not been established in the available materials. Therefore, the paper does not claim that it is a publicly representative financial dataset.

Data Limitations

The records span approximately five calendar months, with incomplete coverage of March. The dataset is relatively small and does not represent a large population of users.

These characteristics limit the conclusions that can be drawn from the experiment. In particular, the dataset is insufficient to establish the reliability of long-term monthly expenditure forecasting.

IV. METHODOLOGY

The proposed workflow consists of data inspection, preprocessing, exploratory analysis, model development, and evaluation.

A. Data Inspection and Preprocessing

The dataset was loaded into Python using the Pandas library. The initial inspection examined the number of records, column names, data types, and missing values.

The preprocessing stage standardized transaction fields, processed transaction dates, and prepared features for analysis. Date-derived information, including month and day-related attributes, was used to explore transaction patterns.

The target variable was standardized as Income or Expense. Records outside these two target classes were excluded from the classification experiment.

Missing transaction descriptions were handled by assigning a placeholder value. Numeric and categorical features were processed through a machine learning preprocessing pipeline.

B. Exploratory Data Analysis

Exploratory data analysis was performed to understand the structure of the financial records.

The analysis included:

1. Examining the distribution of Income and Expense transactions.
2. Studying transaction counts across months.
3. Exploring spending categories.
4. Examining missing values and data types.
5. Generating graphs to visualize transaction patterns.

The dataset contained 35 transactions in November 2021, 89 in December 2021, 92 in January 2022, 54 in February 2022, and 7 in March 2022.

These counts describe the available records; they should not be interpreted as complete monthly expenditure totals or evidence of a general spending trend.

C. Feature Engineering

The classification model used the following input information:

- Transaction amount in Indian rupees.
- Month-related numerical information.
- Day of the month.
- Day of the week.
- Transaction description text.

Numerical features were processed using median imputation and standardization. Day-of-week information was converted into numerical categorical features using one-hot encoding.

Transaction descriptions were represented numerically using TF-IDF with unigram and bigram features.

The original transaction category, account field, and direct Income/Expense label column were excluded from the model inputs to reduce direct target leakage.

D. Logistic Regression

Logistic Regression was selected as the primary classification algorithm. It is a supervised learning method suitable for binary classification.

The model learns a relationship between the input features and the transaction class. For a binary problem, the predicted probability can be represented as:

[
P(y=1\mid x)=\frac{1}{1+e^{-(w^Tx+b)}}
]

where:

- (x) represents the input feature vector.
- (w) represents the learned model weights.
- (b) represents the bias.
- (P(y=1\mid x)) represents the estimated probability of one class.

The model assigns a transaction to a class according to its predicted probability.

The implementation uses class weighting to account for the imbalance between Income and Expense records.

E. Baseline Model

A majority-class baseline was implemented using a classifier that always predicts the most frequent class in the training data.

This baseline provides a simple reference point. A more sophisticated model should be evaluated against it to determine whether it provides an improvement on the held-out test data.

F. Training and Testing

The dataset was divided into training and testing subsets using a 75:25 split. Stratified splitting was used to preserve the approximate class distribution in both subsets.

The random state was set to 42 to support reproducibility.

Both the baseline and Logistic Regression models were evaluated on the held-out test subset.

G. Evaluation Metrics

The following metrics were used.

Accuracy: The proportion of all predictions that are correct.

[
Accuracy=\frac{TP+TN}{TP+TN+FP+FN}
]

Precision: The proportion of positive predictions that are correct.

$$
Precision=\frac{TP}{TP+FP}
$$

Recall: The proportion of actual positive cases correctly identified.

$$
Recall=\frac{TP}{TP+FN}
$$

F1-score: The harmonic mean of precision and recall.

$$
F1=2\frac{Precision\times Recall}{Precision+Recall}
$$

Weighted precision, weighted recall, and weighted F1-score account for the number of test samples belonging to each class.

V. EXPERIMENTAL RESULTS

A. Model Performance

The recorded notebook outputs produced the following comparison:

TABLE I. MODEL PERFORMANCE COMPARISON

Evaluation metric	Majority-class baseline	Logistic Regression
Accuracy	82.9%	94.3%
Weighted precision	68.7%	94.7%
Weighted recall	82.9%	94.3%
Weighted F1-score	75.1%	93.8%

The results show that Logistic Regression achieved higher scores than the majority-class baseline across all four reported metrics.

The accuracy increased from 82.9% to 94.3%, an improvement of 11.4 percentage points on the recorded test split.

The weighted F1-score increased from 75.1% to 93.8%. This indicates that the model performed better overall across the two classes in this particular experiment.

These results apply only to the recorded train-test split. They do not establish that the same performance will be achieved on other datasets or future transactions.

B. Class-Level Evaluation

The recorded classification report provides the following results:

TABLE II. CLASSIFICATION REPORT FOR LOGISTIC REGRESSION

Class	Precision	Recall	F1-score	Test support

Expense	0.94	1.00	0.97	58
Income	1.00	0.67	0.80	12
Macro average	0.97	0.83	0.88	70
Weighted average	0.95	0.94	0.94	70

The model correctly identified all 58 Expense records in the test set. For Income, the recall was 0.67, indicating that approximately two-thirds of the actual Income records were identified correctly.

Although the overall accuracy was high, the lower recall for Income shows that the model did not identify every income transaction. This is important because accuracy alone can hide differences in performance between imbalanced classes.

C. Graphical Results

The notebook generates visualizations for exploratory data analysis and model evaluation.

The transaction distribution graph illustrates the imbalance between Income and Expense records. Monthly transaction-count visualizations show how the number of recorded transactions varies over the available period.

The model comparison graph presents the evaluation metrics for the baseline and Logistic Regression. The confusion matrix provides a class-level view of correct and incorrect predictions.

Figure placement for the final paper:

- **Fig. 1:** Distribution of Income and Expense transactions.

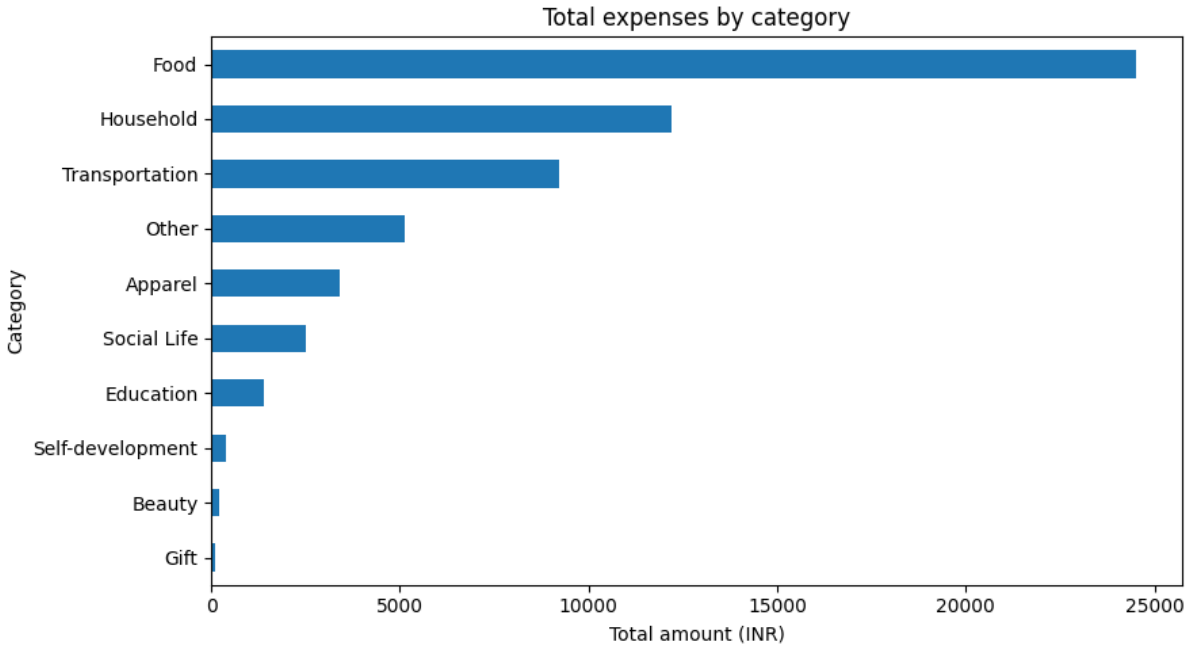


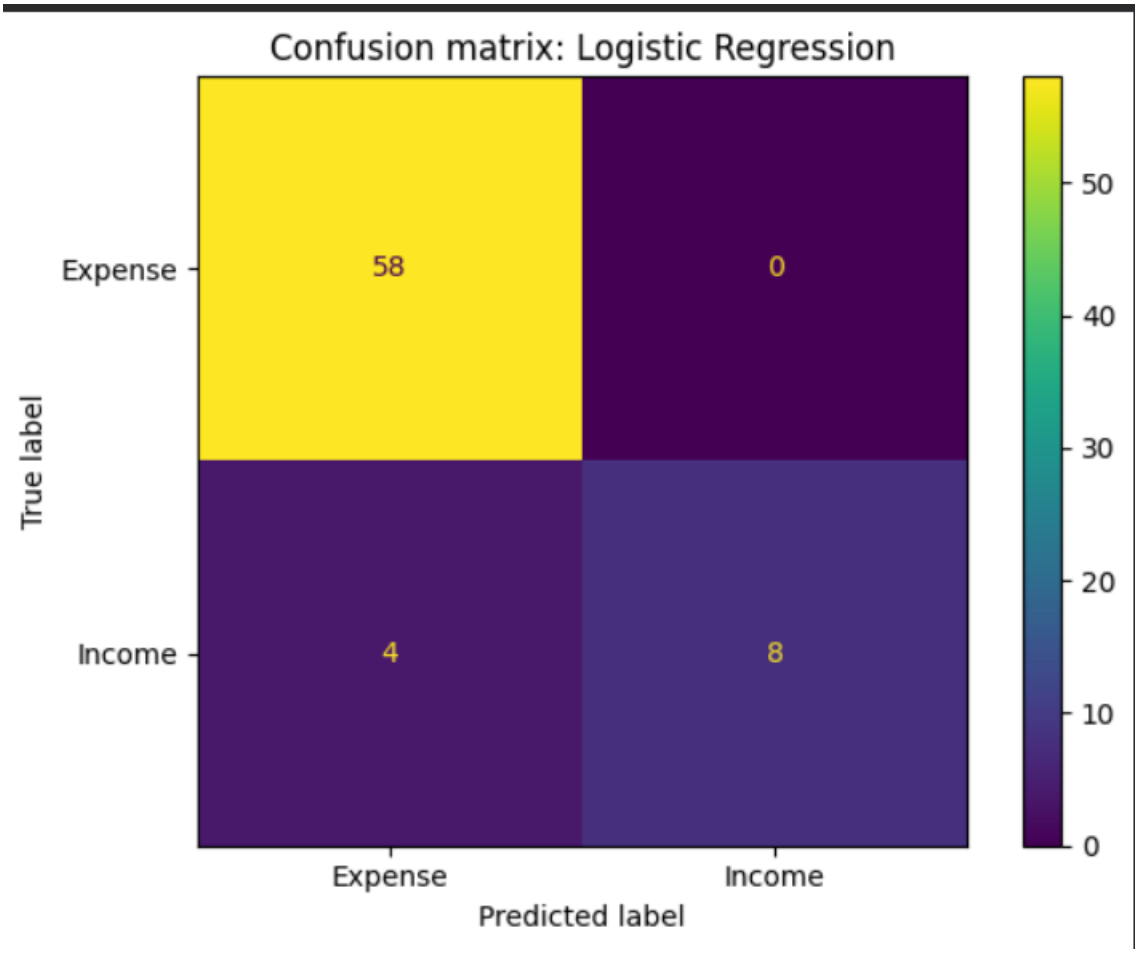
Fig. 2: Monthly transaction counts.

	Amount_INR
count	277.000000
mean	410.750903
std	1065.756569
min	3.000000
25%	50.000000
50%	128.000000
75%	301.150000
max	10000.000000

Fig. 3: Comparison of baseline and Logistic Regression performance.

Model	Accuracy	Weighted precision	Weighted recall	Weighted F1
Majority-class baseline	0.829	0.687	0.829	0.751
Logistic Regression	0.943	0.947	0.943	0.938

- **Fig. 4:** Confusion matrix for Logistic Regression.



VI. DISCUSSION

The experiment demonstrates that a Logistic Regression pipeline combining numerical, categorical, and text features can classify transactions in the supplied dataset more accurately than a majority-class baseline.

The use of TF-IDF allows the model to process information contained in transaction descriptions. Numerical and date-derived features provide additional information that may help distinguish transaction types.

However, the class-level results reveal an important limitation. Income recall was lower than Expense recall. Therefore, the high overall accuracy should not be interpreted as equally strong performance for both classes.

The small dataset and its class imbalance also make the results sensitive to the particular train-test split. A different split could produce different scores.

Further evaluation using repeated stratified cross-validation and an independent dataset would provide stronger evidence of model reliability.

The experiment should therefore be viewed as a preliminary implementation of machine learning for financial transaction classification, rather than a production-ready financial management system.

VII. LIMITATIONS AND FUTURE WORK

The project has several limitations.

1. Limited dataset size: The dataset contains only 277 transactions. A larger dataset would allow more reliable model evaluation.

2. Short observation period: The available records cover approximately five months and have incomplete March coverage. This is insufficient to validate long-term monthly expenditure forecasting.

3. Class imbalance: Expense transactions substantially outnumber Income transactions. Additional evaluation is needed to ensure that the model performs reliably for both classes.

4. Limited generalizability: The results are based on one supplied dataset. The model has not been validated across different users, accounts, or financial circumstances.

5. Limited model comparison: The current implementation compares Logistic Regression with a majority-class baseline. Additional models and repeated evaluation could provide a more comprehensive comparison.

Future work could address these limitations by:

- Collecting a larger and more diverse transaction dataset.
- Evaluating additional classifiers, such as Decision Trees, Random Forests, and Support Vector Machines.
- Using repeated stratified cross-validation.
- Investigating performance separately for Income and Expense.
- Studying category-level spending patterns with suitable data.
- Exploring monthly expenditure forecasting with a sufficiently long, consistently recorded time series.
- Testing the final model on an independent dataset.

Any future forecasting experiment should use a time-based evaluation strategy to avoid training on information that would not have been available at prediction time.

VIII. CONCLUSION

This project implemented an exploratory personal financial transaction analysis and classification workflow using Python and machine learning.

The dataset contained 277 transactions recorded between November 2021 and March 2022. Exploratory data analysis was used to examine transaction distributions and monthly record counts. A Logistic Regression classifier was developed to classify transactions as Income or Expense and was compared with a majority-class baseline.

The recorded Logistic Regression results achieved 94.3% accuracy and a weighted F1-score of 93.8%, compared with 82.9% accuracy and a weighted F1-score of 75.1% for the baseline.

Despite these results, Income recall was lower than Expense recall, and the small, imbalanced dataset limits the generalizability of the findings.

Overall, the experiment demonstrates a practical application of data preprocessing, exploratory analysis, text feature extraction, supervised learning, and model evaluation to personal financial records. Larger datasets and additional validation are necessary before drawing conclusions about performance on future transactions or using the approach for monthly expenditure forecasting.

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Reproducibility note: The implementation is provided in the accompanying Google Colab notebook. The notebook should be saved alongside the dataset and final paper so that the experiment can be inspected and rerun.